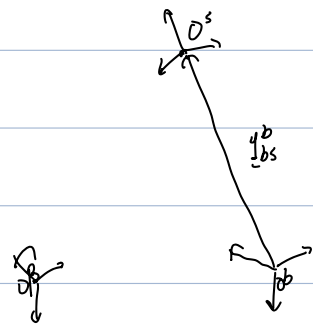


Transposition of Navigation Solution



$p \in \{i, e, n\}$

$$\underline{r}_{ps}^B = \underline{r}_{pb}^B + C_b^B \underline{l}_{bs}^b$$

$$\Rightarrow \dot{\underline{r}}_{ps}^B = \dot{\underline{r}}_{pb}^B + \dot{C}_b^B \underline{l}_{bs}^b + \underbrace{C_b^B \dot{\underline{l}}_{bs}^b}_0 = \underline{V}_{pb}^B + \dot{C}_b^B \underline{l}_{bs}^b$$

resolved in the general frame Y .

$$\begin{aligned} \underline{V}_{ps}^Y &= C_Y^Y \underline{r}_{ps}^B = \underline{V}_{pb}^Y + C_Y^Y \dot{C}_b^B \underline{l}_{bs}^b \\ &= \underline{V}_{pb}^Y + C_Y^Y C_b^B \Omega_{pb}^b \underline{l}_{bs}^b \\ &= \underline{V}_{pb}^Y + C_Y^Y \Omega_{pb}^b \underline{l}_{bs}^b \end{aligned}$$

when $Y = b$ (i.e., resolved in the body frame):

$$\begin{aligned} \underline{V}_{ps}^b &= \underline{V}_{pb}^b + \Omega_{pb}^b \underline{l}_{bs}^b \\ &= \underline{V}_{pb}^b + \underline{\omega}_{pb}^b \times \underline{l}_{bs}^b \end{aligned}$$

Fig. (22, p74)

$$\underline{V}_{es}^b = C_e^b \underline{V}_{es}^e$$

$$\underline{r}_{es}^e = \underline{r}_{eb}^e + C_b^e \underline{l}_{bs}^b$$

$$\begin{aligned} \Rightarrow \underline{V}_{es}^e &= \dot{\underline{r}}_{es}^e = \dot{\underline{r}}_{eb}^e + \dot{C}_b^e \underline{l}_{bs}^b + \underbrace{C_b^e \dot{\underline{l}}_{bs}^b}_0 \\ &= \underline{V}_{eb}^e + C_b^e \Omega_{eb}^b \underline{l}_{bs}^b \end{aligned}$$

$$\begin{aligned} \Rightarrow \underline{V}_{es}^b &= C_e^b \underline{V}_{eb}^e + C_e^b C_b^e \Omega_{eb}^b \underline{l}_{bs}^b \\ &= \underline{V}_{eb}^b + \Omega_{eb}^b \underline{l}_{bs}^b \end{aligned}$$

$$= \underline{u}_b + \Omega_{eb}^b \underline{z}_{bs}$$

$$= \begin{bmatrix} u \\ v \\ w \end{bmatrix} + \begin{bmatrix} 0 & -r_{eb}^b & q_{eb}^b \\ r_{eb}^b & 0 & -p_{eb}^b \\ -q_{eb}^b & p_{eb}^b & 0 \end{bmatrix} \begin{bmatrix} x_{bs}^b \\ y_{bs}^b \\ z_{bs}^b \end{bmatrix}$$

$$= \begin{bmatrix} u - r_{eb}^b y_{bs}^b + q_{eb}^b z_{bs}^b \\ v + r_{eb}^b x_{bs}^b - p_{eb}^b y_{bs}^b \\ w - q_{eb}^b x_{bs}^b + p_{eb}^b y_{bs}^b \end{bmatrix} = \begin{bmatrix} 270.39 \\ -0.295 \\ -0.4 \end{bmatrix}$$